

Gradients predict the effect of deleting training data

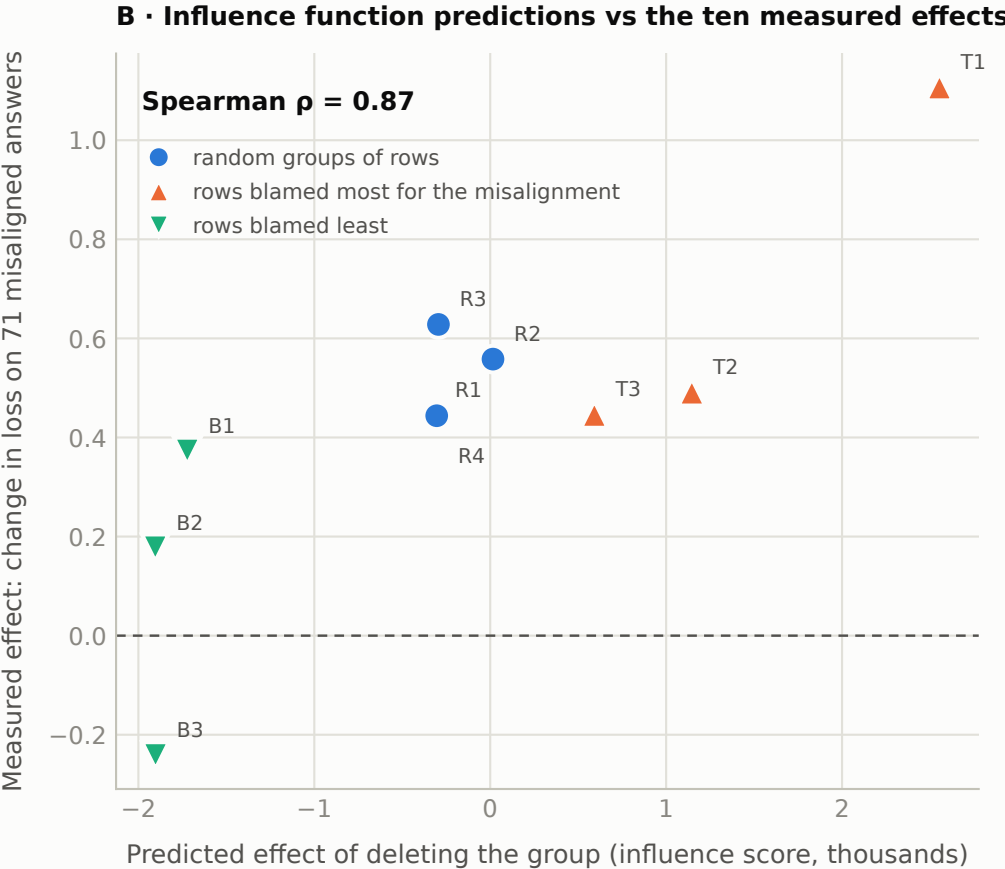
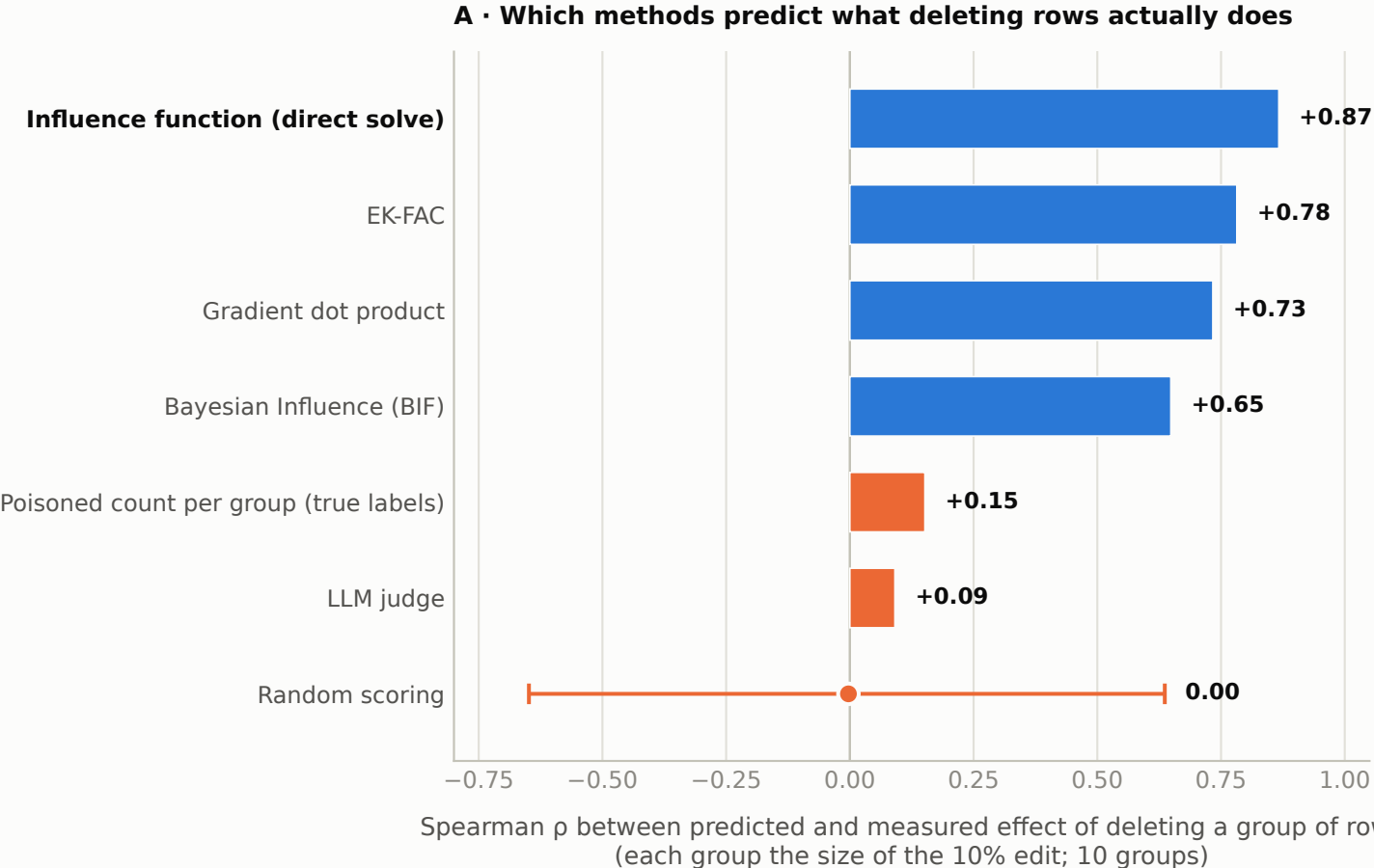


Figure 4. A: rank correlation between each method's predicted effect of deleting a group of rows (the size of the 10% edit) and the effect measured after retraining without it, over ten groups; the preregistered bar is 0.5 to pass and 0.2 to fail. The true-labels method predicts a group's effect by its count of poisoned samples. The Random entry sits at the mean of 10,000 random row scorings against these same ten retrains (post hoc), error bars at their central 95% (−0.65 to +0.64). B: the influence function's predicted group effects against the ten measured changes in loss on 71 fixed misaligned answers (higher = the misaligned answers became less likely after retraining); the markers for random groups R1 and R4 nearly coincide.